

Problem

Given the following state equation, solve for $y(t)$:

$$\dot{\mathbf{x}} = \begin{bmatrix} -4 & 4 \\ -2 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 2 \end{bmatrix} u(t)$$

$$\mathbf{y}(t) = [1 \quad 0] \mathbf{x}$$

Step-by-step solution

Step 1 of 1

We transform the state equations into S – Domain and solve using lap lace transform.

$$SX(S) - X(0) = AX(S) + B\left(\frac{1}{S}\right)$$

Assume initial conditions are Zero

$$(ST - A)X(S) = B\left(\frac{1}{S}\right)$$

$$X(S) = \begin{bmatrix} S+4 & -4 \\ 2 & S \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 2 \end{bmatrix} \left(\frac{1}{S}\right) = \frac{1}{S^2 + 4S + 8} \begin{bmatrix} S & 4 \\ 2 & S+4 \end{bmatrix} \begin{bmatrix} 0 \\ 2/S \end{bmatrix}$$

$$\begin{aligned} Y(S) = X_1(S) &= \frac{8}{S(S^2 + 4S + 8)} = \frac{1}{S} + \frac{-S-4}{S^2 + 4S + 8} \\ &= \frac{1}{S} + \frac{(-S-4)}{(S+2)^2 + 2^2} = \frac{1}{S} + \frac{-(S+4)}{(S+2)^2 + 2^2} + \frac{-2}{(S+2)^2 + 2^2} \end{aligned}$$

$$y(t) = (1 - e^{-2t}(\cos 2t + \sin 2t))u(t)$$